Samyak Kumar

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Professional Summary

Undergraduate student with a strong interest in marine robotics, focused on the design and development of sensor-rich Autonomous Underwater Vehicles (AUVs). Demonstrates systems-level expertise spanning electrical and mechanical integration, embedded electronics, and robotics software, with extensive hands-on experience working with acoustic, inertial, and navigation sensors commonly used in research and experimental AUV platforms, where careful integration and validation are critical. Work centers on the design, integration, and validation of multi-sensor architectures using ROS-based pipelines and state estimation frameworks to develop underwater autonomous systems that perform reliably in real-world environments.

Education

B.Tech in Naval Architecture and Ocean Engineering, Indian Institute of Technology Madras

- **Expected Graduation:** July 2027
- **Coursework:** Marine Robotics, Marine Instrumentation, Maneuvering and Control of Marine Vehicles, Ship Hydrodynamics, Vibration of Marine Structures, Ship Design, Design of Ocean Structures, Marine Engineering

Experience

AUV Sensing and Control Engineer, Team Aritra - IIT Madras, India Dec 2025 - Ongoing

- Selected as one of the teams from India to compete in Singapore AUV Challenge 2026, focusing on full-autonomy underwater missions in a constrained pool environment
- Implementing AUV autonomy for navigation, target detection, and communication-based localization under strict safety and time constraints
- Developing sensor-driven state estimation and control pipelines using IMU, pressure sensor, Echosounder, DVL, imaging sonar, acoustic modem, and underwater GPS
- Implementing depth, heading, and velocity control using multi-sensor fusion to ensure stable motion without touching pool walls, floor, or mission structures
- Designing perception and decision logic for gate detection, obstacle avoidance, and target zone localization using imaging sonar and acoustic cues
- Designed and fabricated a custom ball dropping and pickup mechanism for target acquisition and reacquisition, optimized for reliability, alignment tolerance, and minimal actuation time

Robotics Systems Engineer (AUVs), MAVLAB - IIT Madras, India Aug 2025 - Ongoing

- Designed, built, and iterated on a compact Autonomous Underwater Vehicle (AUV), owning the electrical, mechanical, and autonomy stack from component selection to pool-validated operation for shallow-water surveys
- Architected the electrical system including power distribution, sensor and thruster interfacing, grounding strategy, and internal wiring layout to ensure reliability during submerged operation
- Integrated a heterogeneous bathymetric sensing suite including pressure sensor, DVL, Echosounder, Imaging sonar, Side-scan sonar, Forward-looking imaging sonar, and IMU, addressing physical placement, electrical interfacing, and timing synchronization to ensure consistent depth and acoustic range measurements
- Developed a ROS-based data acquisition and logging architecture for synchronized collection of depth, inertial, velocity, and acoustic data, enabling repeatable bathymetric survey experiments
- Implemented EKF-based navigation and depth estimation using IMU, DVL, and pressure sensor data to provide stable vehicle state estimates required for geo-referenced bathymetric mapping
- Developing a custom sonar data visualization and interaction interface to ingest synchronized outputs from front-scan sonar, side-scan sonar, and Imaging sonar, enabling interactive inspection and interpretation of acoustic returns for bathymetric analysis

Electrical and Mechanical Design Specialist, Team Amogh - IIT Madras, India Mar 2025 - Nov 2025

- Core member of Team Amogh, contributing to the design and development of an Autonomous Underwater Vehicle (AUV) for the RoboSub 2026 competition
- Designed and implemented the electrical architecture of the AUV, including PCB layouts, power regulation, sensor and thruster interconnections, and system-level grounding for operational safety
- Developed schematics and PCB layouts using KiCad, optimizing component density, power distribution, and electrical reliability
- Executed hardware integration and assembly, including waterproofing, wiring, soldering, and mechanical mounting of internal components within the pressure housing
- Developed key aspects of the mechanical system design, including thruster placement, hydrodynamic considerations, and 3D CAD modeling using Fusion 360 to improve vehicle stability and reduce drag

Technical Skills

AUV Autonomy, Estimation & Control

- Autonomous mission execution for AUVs in constrained pool environments
- EKF-based navigation and depth estimation using IMU, DVL, pressure sensor, and acoustic sensors
- Depth, heading, and velocity control through multi-sensor fusion
- Stability-focused control design under strict safety and boundary constraints

Underwater Perception & Multi-Sensor Software

- Developed ROS 2–based architectures for multi-sensor data acquisition, integration, and logging
- Imaging, forward-looking, and side-scan sonar processing for perception and localization
- Sonar data preprocessing, representation, and interpretation for underwater environments
- Custom sonar visualization and interaction tools for inspection and validation
- Python for robotics algorithm development, sensor data processing, and analysis

Electrical & Mechanical Systems Integration

- Electrical system architecture for AUVs, including power distribution, grounding, and safety systems
- PCB schematic capture and layout using KiCad
- Mechanical design, internal layout, and integration using Fusion 360
- Hands-on fabrication including wiring, soldering, waterproofing, and pressure-housing integration
- Electrical interfacing and system-level integration of sensors, thrusters, and actuators within sealed AUV pressure housings

Conference Attended

- **OCEANS Conference - Great Lakes , Chicago, USA**
  Presented student section initiatives, interacted with global researchers and AUV experts in marine technology
- **International Conference in Ocean Engineering (ICOE) 2025, Chennai, India**
  Attended technical sessions on marine hydrodynamics, ocean observation systems, and marine robotics

Research Interests

•Underwater Robotics • Acoustic sensing • Sonar-based perception • System-level AUV design and integration

Awards and Achievements

- **MTS Dieter Family Scholarship, Marine Technology Society - 2025**
  Awarded for academic merit and active contribution to student marine technology initiatives

Leadership and Extracurricular Activities

- **Joint Secretary, Marine Technology Society (MTS) IIT Madras Student Chapter:**
  Coordinated Techsym 2025 and Aquavision AUV competition; managed collaborations with NIOT and IEEE OES
- **Student Ambassador, Royal Institute of Naval Architects (RINA):**
  Representing IIT Madras in professional networks through RINA initiatives and promoting marine technology